

Exploratory Data Analysis of

Netflix Movies Dataset

A statistical analysis of 9,827 movie records covering genre, popularity, and audience ratings

9,827 Movies Analyzed	19 Unique Genres	1902–2024 Years Covered	0 Missing/Duplicate Records
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1. Introduction

This report presents an exploratory data analysis (EDA) of a Netflix-style movie dataset containing 9,827 records sourced from The Movie Database (TMDB). The dataset captures each title's release date, popularity score, vote count, average audience rating, genre classification(s), and supporting metadata. The goal is to understand the catalogue's genre composition, characterize how popularity and ratings are distributed, and identify patterns in release activity over time.

The raw dataset contains 9 columns. For this analysis, the Overview, Original_Language, and Poster_Url fields were dropped, as they do not contribute to the genre, popularity, or rating analysis.

2. Data Preparation

The dataset was clean on arrival: no missing values and no duplicate rows. Preparation therefore focused on type correction and feature engineering:

- Vote_Count and Vote_Average were coerced to numeric types for correct aggregation and statistics.
- Release_Date was parsed into a datetime, then reduced to release year only, yielding a usable range of 1902–2024.
- Vote_Average was bucketed into four equal-frequency categories — Not Popular, Below Average, Average, and Popular — using quartile cut points.
- The Genre field stored multiple comma-separated genres per movie (e.g. “Action, Adventure, Science Fiction”). This was split and exploded so each movie-genre pair occupies its own row, expanding the table from 9,827 to 25,793 rows for genre-level analysis.

3. Key Findings

3.1 Genre Distribution

Drama is the most frequent genre, appearing in 3,744 of the 25,793 movie-genre entries (14.5%). Comedy follows with 3,031 entries (11.8%), and Action ranks third with 2,686 (10.4%). Together, Drama, Comedy, Action, and Thriller account for nearly half of all genre tags, while niche genres such as Western (137), TV Movie (214), and Documentary (215) each make up less than 1%.

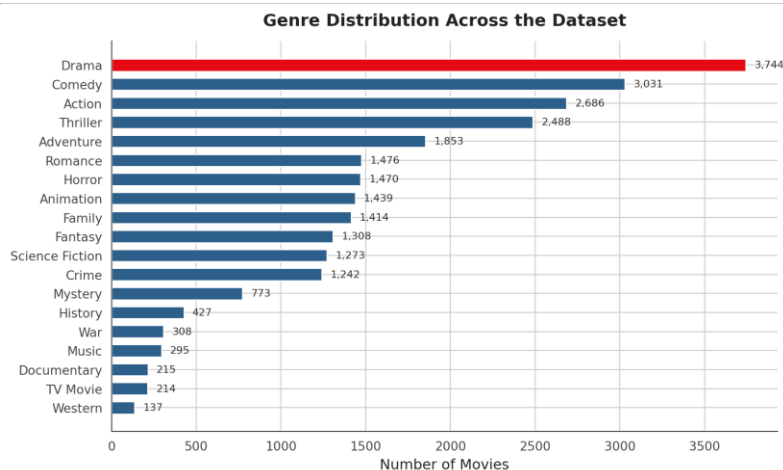


Figure 1. Number of movies tagged with each genre (a movie may carry multiple genre tags).

3.2 Popularity and Vote Distribution

Audience ratings, once bucketed, are nearly evenly split: Average (6,613, 25.6%), Popular (6,520, 25.3%), Below Average (6,348, 24.6%), and Not Popular (6,071, 23.5%) — expected since the bands were built from quartiles, confirming no unusual skew in the underlying ratings. Within the “Popular” category specifically, Drama again leads, accounting for roughly one in five (about 20%) of all popular-rated entries.

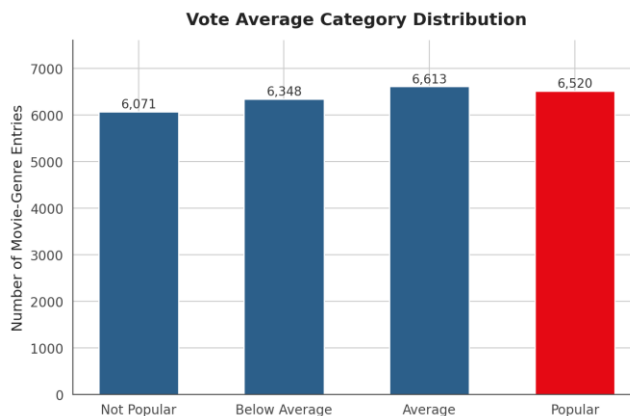


Figure 2. Distribution of movie-genre entries across the four Vote Average categories.

3.3 Most and Least Popular Titles

Spider-Man: No Way Home (2021) is the standout by popularity score (5,083.95) — over four times the next-highest title. It carries the genres Action, Adventure, and Science Fiction, with a strong 8.3 rating from 8,940 votes, showing its popularity is backed by genuine reception rather than novelty alone. At the opposite end, two titles tie for the lowest score (13.35): The United States vs. Billie Holiday (2021) and Threads (1984) — both lower-profile, drama-driven films rather than blockbusters.

3.4 Release Activity Over Time

Release volume rises steadily from the 1960s and accelerates sharply after 2010. The busiest year is 2021, with 714 titles — nearly double 2015's count (390). The drop after 2021 reflects the dataset's collection cutoff, not an actual decline in output.

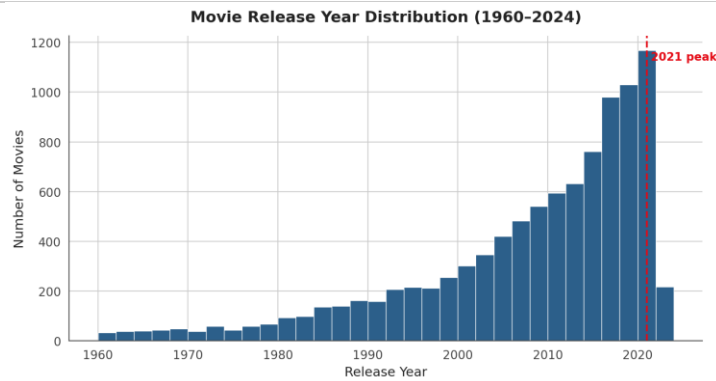


Figure 3. Count of movies by release year, 1960–2024.

Tracking the top five genres since 2015, Drama consistently leads in volume and peaks sharply in 2021, alongside a broad rise across Comedy, Action, and Thriller that same year — consistent with a general surge in releases rather than a genre-specific shift. Adventure stays comparatively flat across the period.

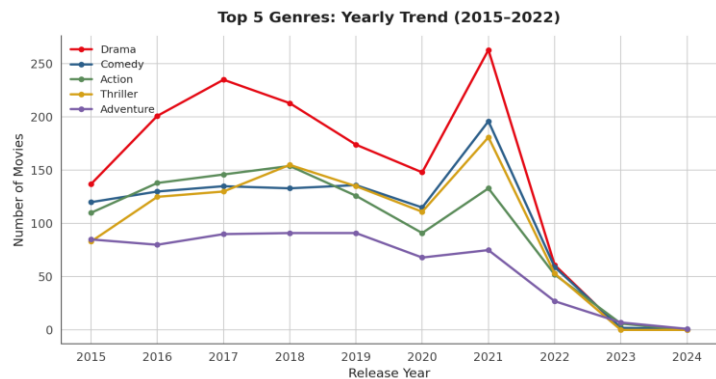


Figure 4. Yearly release counts for the five most common genres, 2015–2022.

4. Conclusion

This analysis confirms Drama as the defining genre of the dataset — both in raw volume and in audience-rated popularity — while Comedy and Action form a strong second tier. Ratings show no significant skew once bucketed, and release activity is heavily concentrated in the post-2015 period, peaking in 2021; earlier decades are comparatively under-represented, which matters for time-based trend analysis. Overall, the dataset offers a reliable basis for genre-level and popularity-based content analysis, with the main caveats being its recency bias and the fact that a single movie may be represented across multiple genre categories.

Summary of Findings

- Drama is the most frequent genre (3,744 entries, 14.5%), followed by Comedy (3,031, 11.8%) and Action (2,686, 10.4%).
- Rating categories are nearly evenly distributed; Drama leads the “Popular” category with roughly 20% of those entries.
- Spider-Man: No Way Home (2021) holds the highest popularity score (5,083.95), backed by an 8.3 rating from 8,940 votes.
- 2021 is the peak release year, with 714 titles — nearly double the 2015 count.
- The dataset is clean (no missing values, no duplicates) but skews heavily toward 2015–2022 releases.